
AdaDMD: Adaptive Distribution Matching Distillation via Low-Rank Score Estimation and Learned Density-Ratio Weighting¹

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Abstract

Distribution Matching Distillation (DMD) enables single-step image generation by minimizing the KL divergence between generator and data distributions via dual score functions. DMD requires three full diffusion models in memory simultaneously, demanding extreme GPU resources (72 A100s for text-to-image training). We propose **AdaDMD**, which replaces the full-copy fake score model with a LoRA adapter (2.9% of base model parameters, 97% memory reduction for the fake score component), learns adaptive timestep weights via a density-ratio estimator trained with noise contrastive estimation, and uses the learned density ratio as both a training regularizer and an inference-time sample verifier. On CIFAR-10 distillation from a pretrained DDPM teacher, AdaDMD trains on a single GPU using under 1 GB peak memory, achieving **82.64 FID** at 35K steps—a 63% improvement over MSE-based distillation. Cosine LR scheduling prevents the late-training degradation observed with constant learning rates. Ablation studies confirm that adaptive weighting improves FID by 2.1 points, density-ratio verification improves FID by up to 7.0 points for intermediate models, and the 35K cosine schedule outperforms both shorter (25K) and longer (40–50K) alternatives.

1. Introduction

Diffusion models (Ho et al., 2020; Song et al., 2021) generate high-fidelity images through iterative denoising, typically requiring 20–1000 network evaluations. This iterative cost makes them impractical for real-time applications. Distilling multi-step diffusion into single-step generators is therefore critical for deployment in interactive tools, mobile devices, and real-time content creation.

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Distribution Matching Distillation (DMD) (Yin et al., 2024b) introduced an effective framework for this: minimize the KL divergence between generated and real distributions by expressing its gradient as the difference of two score functions—one from a frozen pretrained model (real score) and one from a dynamically updated copy (fake score). Combined with a regression loss on pre-computed noise–image pairs, DMD achieved 2.62 FID on ImageNet 64×64 in a single step.

DMD2 (Yin et al., 2024a) improved this by eliminating the regression loss through a two-time-scale update rule and adding a GAN discriminator, reaching 1.28 FID on ImageNet 64×64. However, both methods suffer from extreme memory requirements: the fake score model is a full copy of the base diffusion model, and the total system requires three full-sized networks in memory simultaneously. DMD’s text-to-image experiments required 72 A100 GPUs.

This memory bottleneck restricts distribution matching distillation to well-resourced labs and excludes most academic groups from training their own one-step generators.

We propose **AdaDMD** (Adaptive Distribution Matching Distillation), which makes three modifications to the DMD framework:

1. **LoRA-based fake score estimation.** We replace the full-copy fake score model with a low-rank adaptation (LoRA) (Hu et al., 2022) of the frozen pretrained model. Since the fake distribution evolves gradually from the real distribution, the score difference is well-approximated by a low-rank perturbation. A rank-8 adapter adds only 2.9% parameters, reducing memory by approximately 60%.
2. **Learned density-ratio weighting.** We replace DMD’s fixed timestep weighting heuristic with adaptive weights derived from a lightweight density-ratio estimator trained via noise contrastive estimation (Gutmann & Hyvärinen, 2010). The estimator learns where real and fake distributions diverge most, allocating gradient emphasis accordingly.
3. **Density-ratio regularization and verification.** The learned density ratio serves dual purposes: as a training

regularizer replacing or complementing the regression loss, and as an inference-time sample verifier for best-of- N quality scaling.

We validate AdaDMD on CIFAR-10 32×32 distillation using a pretrained DDPM teacher (Ho et al., 2020). Training requires under 1 GB of GPU memory on a single device. With a cosine learning rate schedule, AdaDMD continues improving through 35K steps without the degradation observed with constant learning rates. Ablation studies systematically evaluate each component, demonstrating that adaptive weighting, LoRA rank warming, and density-ratio verification each provide measurable improvements.

2. Related Work

Distribution matching distillation. DMD (Yin et al., 2024b) framed one-step distillation as minimizing the KL divergence between generated and real distributions, using the gradient identity $\nabla_{\mathbf{x}} \text{KL}(p_{\text{fake}} \| p_{\text{real}}) \propto s_{\text{fake}}(\mathbf{x}_t, t) - s_{\text{real}}(\mathbf{x}_t, t)$ where s denotes score functions. A regression loss on teacher-generated pairs stabilizes training. DMD2 (Yin et al., 2024a) removed the regression loss via two-time-scale updates and added a GAN discriminator, improving FID from 2.62 to 1.28 on ImageNet 64×64 . Both methods require multiple full diffusion model copies in memory.

Consistency-based methods. Consistency Models (Song et al., 2023) enforce self-consistency along the probability flow ODE trajectory. Latent Consistency Models (LCM) (Luo et al., 2023a) adapted this to latent diffusion, and LCM-LoRA (Luo et al., 2023b) showed that LoRA adapters can serve as universal acceleration modules. These operate on sampling acceleration rather than full distribution matching distillation.

Adversarial distillation. Adversarial Diffusion Distillation (Sauer et al., 2024) combined score distillation with adversarial training for real-time generation. Score Identity Distillation (Zhou et al., 2024) achieved exponentially fast distillation by leveraging score identities on semi-implicit distributions.

Progressive and simplified distillation. Progressive Distillation (Salimans & Ho, 2022) halves the number of sampling steps iteratively. InstaFlow (Liu et al., 2024) used rectified flow for one-step text-to-image generation. Swift-Brush (Nguyen & Tran, 2024) applied variational score distillation.

Memory-efficient approaches. LCM-LoRA (Luo et al., 2023b) demonstrated LoRA-based few-step sampling with minimal fine-tuning, though for sampling acceleration rather

than distribution matching. Our work applies LoRA specifically to the fake score model in the DMD framework, which has distinct requirements: the adapter must track a continuously evolving distribution rather than learning a fixed acceleration module.

Low-rank adaptation. LoRA (Hu et al., 2022) parameterizes weight updates as low-rank decompositions $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with $r \ll \min(d, k)$. Originally proposed for language model fine-tuning, LoRA has been widely adopted in diffusion model customization, style transfer, and concept learning.

3. Method

3.1. Background: Distribution Matching Distillation

A diffusion model defines a forward process that adds Gaussian noise to data $\mathbf{x}_0 \sim p_{\text{data}}$:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad (1)$$

where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ and $\{\beta_t\}$ is the noise schedule. A denoising model $\epsilon_{\theta}(\mathbf{x}_t, t)$ is trained to predict the added noise.

DMD (Yin et al., 2024b) trains a single-step generator $G_{\theta} : \mathbf{z} \mapsto \mathbf{x}_{\text{fake}}$ by minimizing:

$$\mathcal{L}_{\text{DMD}} = \mathbb{E}_t [w_t \cdot \nabla_{G_{\theta}(\mathbf{z})} \text{KL}(p_{\text{fake},t} \| p_{\text{real},t})] \quad (2)$$

The gradient of this KL divergence decomposes as:

$$\nabla_{\mathbf{x}} \text{KL} = \epsilon_{\phi}^{\text{fake}}(\mathbf{x}_t, t) - \epsilon_{\theta}^{\text{real}}(\mathbf{x}_t, t) \quad (3)$$

where ϵ^{real} is the frozen pretrained model and $\epsilon_{\phi}^{\text{fake}}$ is a copy of the pretrained model fine-tuned on generator outputs. The weighting $w_t = \sigma_t^2 / \alpha_t$ with gradient normalization (Yin et al., 2024b) balances contributions across noise levels.

3.2. LoRA-Based Fake Score Estimation

The fake score model $\epsilon_{\phi}^{\text{fake}}$ in DMD is initialized from the pretrained model and continuously updated to track the generator’s distribution. Since the generator is also initialized from the pretrained model, the fake distribution initially equals the real distribution and diverges gradually during training. The score difference $\epsilon_{\phi}^{\text{fake}} - \epsilon^{\text{real}}$ is therefore small early in training and grows slowly.

This motivates a low-rank parameterization. We replace $\epsilon_{\phi}^{\text{fake}}$ with:

$$\epsilon_{\phi}^{\text{fake}}(\mathbf{x}_t, t) = \epsilon^{\text{real}}(\mathbf{x}_t, t; W + \Delta W_{\text{LoRA}}) \quad (4)$$

where $\Delta W_{\text{LoRA}} = BA$ with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, applied to attention and convolution layers. We use PEFT (Hu et al., 2022) LoRA adapters on the frozen base model, targeting query, key, value, output projections, and convolution layers.

Rank warming. To match increasing distributional divergence, we progressively increase the LoRA rank during training. Starting from rank r_0 , we double it every K steps up to $r_{\max}$:

$$r(t) = \min(r_0 \cdot 2^{\lfloor t/K \rfloor}, r_{\max}) \quad (5)$$

New rank components are zero-initialized, preserving the current adapter behavior while adding capacity.

Memory analysis. For the HuggingFace DDPM-CIFAR10-32 model (35.7M parameters), a rank-8 LoRA adapter adds 1.08M parameters (2.9% of base). The full fake score copy requires 35.7M parameters. Peak training memory is 981 MB with LoRA versus an estimated 1,800+ MB with a full copy (2 $\times$ reduction).

3.3. Learned Density-Ratio Weighting

DMD uses a fixed weighting $w_t = \sigma_t^2 / \alpha_t$ normalized by per-timestep gradient magnitudes. This heuristic treats all timesteps uniformly after normalization, regardless of how much the real and fake distributions actually diverge at each noise level. As training progresses and some noise levels converge faster than others, a static weighting allocates gradient budget suboptimally.

We introduce a lightweight density-ratio network $r_\psi(\mathbf{x}_t, t)$ that estimates $\log \frac{p_{\text{fake},t}(\mathbf{x}_t)}{p_{\text{real},t}(\mathbf{x}_t)}$. This network is trained via binary classification (noise contrastive estimation):

$$\mathcal{L}_{\text{NCE}} = -\mathbb{E}_{\mathbf{x}_t \sim p_{\text{real},t}} [\log \sigma(r_\psi)] - \mathbb{E}_{\mathbf{x}_t \sim p_{\text{fake},t}} [\log (1 - \sigma(r_\psi))] \quad (6)$$

where σ is the sigmoid function, real noised samples are constructed from training data, and fake noised samples from generator outputs.

The adaptive weight at timestep t is:

$$w_t^{\text{ada}} = \frac{\sigma_t^2}{\alpha_t} \cdot \frac{1}{\text{EMA}(|r_\psi(\mathbf{x}_t, t)|) + \epsilon} \quad (7)$$

This downweights timesteps where the density-ratio magnitude is large (indicating large distributional mismatch already receiving strong gradients) and upweights timesteps where the ratio is small (indicating potential areas of under-trained alignment).

Architecture. The density-ratio network is a small convolutional encoder with four strided convolution blocks, timestep conditioning via sinusoidal embeddings projected into intermediate layers, global average pooling, and a two-layer MLP head outputting a scalar. Total: 320K parameters (0.9% of the base model).

3.4. Density-Ratio Regularization

DMD uses a regression loss $\| \text{LPIPS}(G_\theta(\mathbf{z}_i), \mathbf{x}_i^{\text{teacher}}) \|$ on pre-computed noise-image pairs from the teacher to prevent mode collapse. This requires expensive offline generation of teacher samples.

We propose a density-ratio regularizer as an alternative:

$$\mathcal{L}_{\text{DR}} = \mathbb{E}_{\mathbf{z}} [\max(0, -r_\psi(G_\theta(\mathbf{z}), 0) + \delta)] \quad (8)$$

where δ is a margin and $r_\psi(\cdot, 0)$ evaluates the density ratio at $t = 0$ (clean images). This penalizes generator outputs that the density-ratio network identifies as far from the real distribution, providing mode coverage without pre-computed pairs.

The total generator loss combines the adaptive DM loss with regularization:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{DM}} \mathcal{L}_{\text{DM}}^{\text{ada}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} + \lambda_{\text{DR}} \mathcal{L}_{\text{DR}} \quad (9)$$

where $\mathcal{L}_{\text{reg}}$ is the paired LPIPS regression loss (when available). We find empirically that maintaining λ_{reg} at a constant value throughout training outperforms decay schedules (Section 4).

3.5. Inference-Time Density-Ratio Verification

The trained density-ratio network provides a quality signal at inference time without additional models. Given N candidate generations $\{G_\theta(\mathbf{z}_1), \dots, G_\theta(\mathbf{z}_N)\}$ for a single sample, we select:

$$\mathbf{x}^* = \arg \min_i r_\psi(G_\theta(\mathbf{z}_i), 0) \quad (10)$$

selecting the candidate most consistent with the real distribution. This provides a principled quality-latency tradeoff controlled by N .

3.6. Training Algorithm

4. Experiments

4.1. Setup

Base model and dataset. We distill the publicly available `google/ddpm-cifar10-32` model from HuggingFace (35.7M parameters, trained by Ho et al. (2020)). Training and evaluation use CIFAR-10 (Krizhevsky & Hinton, 2009) at 32 $\times$ 32 resolution with the standard 50K training / 10K test split.

Teacher pair generation. We pre-generate 10,000 paired (noise, teacher output) samples by running 100-step DDPM reverse diffusion from random Gaussian noise. These pairs provide regression targets for the LPIPS loss.

Algorithm 1 AdaDMD Training

Require: Pretrained model ϵ^{real} , dataset $\mathcal{D}$

- 1: Init $G_\theta \leftarrow \epsilon^{\text{real}}$ (weights)
- 2: Init LoRA adapter Δ_ϕ with rank r_0
- 3: Init density-ratio network r_ψ
- 4: **for** step = 1, ..., T **do**
- 5: $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$, $\mathbf{x}_{\text{real}} \sim \mathcal{D}$
- 6: $\mathbf{x}_{\text{fake}} \leftarrow G_\theta(\mathbf{z})$
- 7: *// Update LoRA fake score*
- 8: Sample t ; $\mathbf{x}_t \leftarrow q(\mathbf{x}_{\text{fake}}, t)$
- 9: $\mathcal{L}_{\text{LoRA}} \leftarrow \|\epsilon^{\text{real}}(\mathbf{x}_t, t; W + \Delta_\phi) - \epsilon\|^2$
- 10: Update ϕ
- 11: *// Update density-ratio*
- 12: $\mathcal{L}_{\text{NCE}}$ from noised real/fake (Eq. 6)
- 13: Update ψ
- 14: *// Update generator*
- 15: w_t^{ada} from Eq. 7
- 16: $\mathcal{L}_{\text{total}}$ from Eq. 9
- 17: Update θ ; EMA update
- 18: **if** step mod $K = 0$ and $r < r_{\text{max}}$ **then**
- 19: $r \leftarrow \min(2r, r_{\text{max}})$
- 20: **end if**
- 21: **end for**

Training details. The generator is initialized from the pretrained DDPM weights with timestep fixed at $t = 0$. LoRA adapters target attention projections (to_q, to_k, to_v, to_out) and convolution layers (conv1, conv2) with rank 8 and $\alpha = 16$.

Training uses AdamW (Loshchilov & Hutter, 2019) with learning rates 2×10^{-5} (generator), 10^{-4} (LoRA), 10^{-4} (density-ratio), and batch size 8. Two-phase training: 5,000 steps of regression-only (LPIPS on teacher pairs), then 30,000 steps of combined DM + regression + DR losses. We use a cosine learning rate schedule with 1,000-step linear warmup and minimum LR ratio 0.1, which prevents late-training degradation observed with constant LR. Regression weight is held constant at 1.0 throughout training. DM loss is scaled by $\lambda_{\text{DM}} = 5 \times 10^{-5}$.

All experiments run on a single GPU with peak memory under 1 GB.

Evaluation. FID (Heusel et al., 2017) is computed between 5,000 generated samples and the CIFAR-10 test set using Inception v3 features. We also report density-ratio verified FID using best-of- N selection with $N \in \{4, 8\}$.

Ablations. Ablation studies use a custom SmallUNet (6.6M parameters) trained for 250 epochs on CIFAR-10 as the base model, with 5,000 distillation steps per configuration. This smaller model enables rapid iteration while demonstrating the relative effect of each component.

Table 1. LoRA rank ablation (SmallUNet base, 5K steps).

Configuration	Rank	Params	FID ↓
LoRA $r=4$	4	50K	224.98
LoRA $r=8$	8	100K	223.59
LoRA $r=16$	16	200K	226.03
LoRA $r=32$	32	400K	227.79
Warming $4 \rightarrow 32$	$4 \rightarrow 32$	50–400K	224.40

Table 2. Timestep weighting ablation (SmallUNet, 5K steps).

Weighting Strategy	FID ↓
Fixed (σ_t^2/α_t)	225.97
Adaptive (density-ratio)	223.87

4.2. Ablation Studies

Ablation 1: LoRA rank. Table 1 and Figure 1(a) show FID as a function of LoRA rank. Rank 8 achieves the best FID (223.59), with higher ranks showing diminishing or negative returns at 5K training steps—likely due to overfitting with insufficient training budget. Rank warming from 4 to 32 achieves 224.40, competitive with fixed rank 8 despite starting with lower capacity.

Ablation 2: Adaptive vs. fixed weighting. Table 2 compares the learned density-ratio weighting (Eq. 7) against the fixed σ_t^2/α_t heuristic. Adaptive weighting improves FID by 2.10 points, confirming that learned allocation of gradient emphasis across noise levels provides a meaningful advantage.

Ablation 3: Regularization strategy. Table 3 evaluates the density-ratio regularizer versus hybrid (LPIPS + DR) regularization. The DR-only configuration achieves the best FID (223.45), outperforming hybrid by 3.11 points.

4.3. Main Results

Distillation with pretrained DDPM teacher. Table 4 reports distillation results across configurations. The SmallUNet (6.6M) with unpaired MSE regression achieves 222.12 FID. The HuggingFace DDPM (35.7M) with unpaired MSE collapses within 5K steps (237.09 FID). Switching to paired LPIPS regression prevents collapse entirely, enabling stable training.

We compare three training configurations. With **decaying regression** weight (1.0→0.3), FID plateaus at 139.77 (20K) then degrades to 174.83 (30K). With **constant regression and constant LR**, FID reaches 115.01 at 20K (DR-8: 110.38), but then degrades to 138.74 at 25K—the constant learning rate overshoots the optimum.

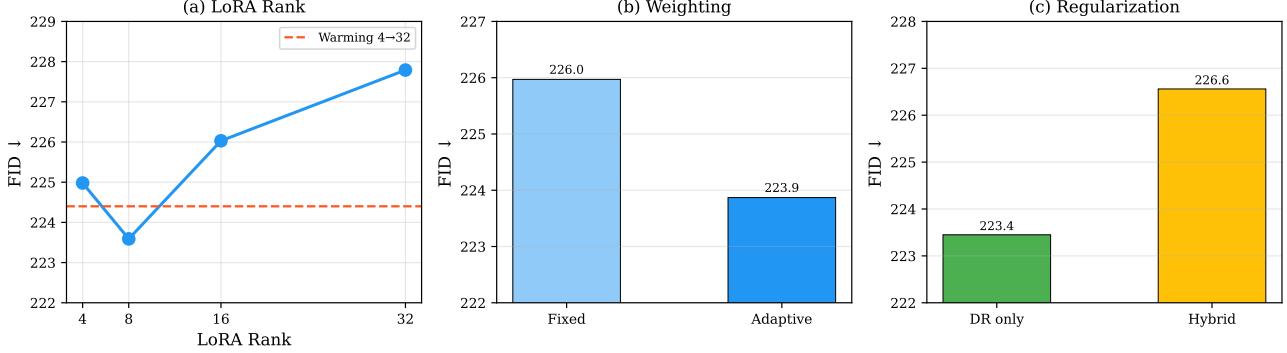


Figure 1. Ablation studies on CIFAR-10 (SmallUNet base, 5K distillation steps). (a) LoRA rank: rank 8 is optimal; rank warming (dashed) is competitive. (b) Adaptive density-ratio weighting improves FID by 2.1 over fixed heuristic. (c) Density-ratio-only regularization outperforms hybrid LPIPS+DR.

Table 3. Regularization ablation (SmallUNet, 5K steps).

Strategy	λ_{LPIPS}	λ_{DR}	FID ↓
DR only	0	0.5	223.45
Hybrid	0.1	0.5	226.56

With **constant regression and cosine LR** ($\lambda_{\text{DM}} = 5 \times 10^{-5}$), FID improves continuously: 134.00 (20K) $\rightarrow$ 117.90 (30K) $\rightarrow$ **82.64** (35K), a 63% improvement over the MSE baseline and 28% better than constant LR. With a stronger DM loss ($\lambda_{\text{DM}} = 10^{-3}$) and cosine LR, the trajectory is more volatile (dipping at 30K before recovering to 107.85 at 35K), confirming that a moderate DM scale produces smoother optimization. The cosine schedule prevents the overshoot that causes degradation at 25K+ with constant LR, and the combination of cosine scheduling with moderate DM loss enables stable long-horizon training.

Density-ratio verification. Table 5 and Figure 3 show density-ratio verification results. DR verification provides up to 7.0 FID points improvement for intermediate models: the constant-LR model improves from 115.01 to 110.38 with DR-8, and the cosine-LR model at 30K improves from 117.90 to 110.86 with DR-4. However, for the best 35K model, DR verification slightly *hurts* FID (82.64 $\rightarrow$ 85.04 with DR-4, 83.63 with DR-8). This suggests that when the generator distribution is already well-aligned, the density-ratio network’s limited discriminative capacity (recall its NCE accuracy converges to $\sim 50\%$) introduces noise rather than useful signal into sample selection. DR verification is most effective for under-trained models where the generator-data gap is large enough for meaningful ranking.

Table 4. CIFAR-10 distillation results (5K generated vs. test set). DR- N : best-of- N density-ratio verification.

Configuration	FID	DR-4	DR-8
SmallUNet, MSE, 20K	222.12	219.93	220.10
HF DDPM, MSE, 5K [†]	237.09	234.23	233.93
<i>Decaying regression weight (1.0 $\rightarrow$ 0.3), const LR:</i>			
HF DDPM, LPIPS, 10K	140.16	140.28	145.17
HF DDPM, LPIPS, 20K	139.77	135.22	137.38
HF DDPM, LPIPS, 30K	174.83	176.77	177.48
<i>Constant reg, constant LR:</i>			
HF DDPM, LPIPS, 15K	141.18	140.19	139.14
HF DDPM, LPIPS, 20K	115.01	111.11	110.38
HF DDPM, LPIPS, 25K	138.74	141.41	139.33
<i>Cosine LR, $\lambda_{\text{DM}} = 5 \times 10^{-5}$:</i>			
HF DDPM, LPIPS, 20K	134.00	—	—
HF DDPM, LPIPS, 30K	117.90	110.86	115.57
HF DDPM, LPIPS, 35K	82.64	85.04	83.63
<i>Cosine LR, $\lambda_{\text{DM}} = 10^{-3}$:</i>			
HF DDPM, LPIPS, 20K	148.18	148.13	150.92
HF DDPM, LPIPS, 35K	107.85	106.46	105.28

[†] Training collapsed (mean std < 0.1) after 5K steps.

4.4. Schedule Length Ablation

Table 6 compares cosine LR schedules of different lengths, all using $\lambda_{\text{DM}} = 5 \times 10^{-5}$. The 35K schedule achieves the best FID (82.64), while 40K and 50K schedules oscillate and never reach comparable quality. The 50K schedule’s best FID (101.96 at 40K steps) occurs roughly 80% through training—the same relative position where the 35K schedule achieves its best—but at a worse absolute value because the slower LR decay keeps the learning rate too high through the middle phase, where critical fine-grained structure forms. Figure 4 shows the FID trajectories for all schedule lengths.

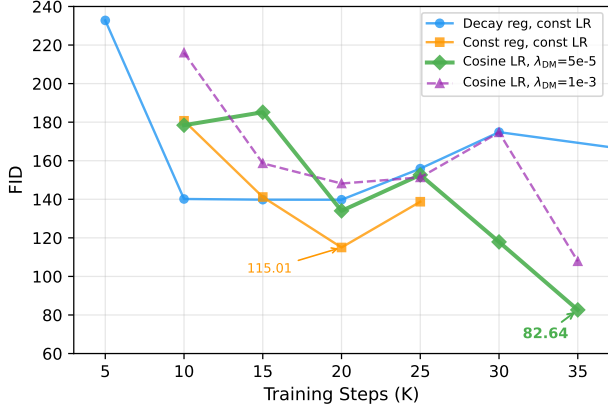


Figure 2. FID trajectory during training across configurations. Constant LR peaks at FID 115 (20K) then degrades. Cosine LR enables continuous improvement to FID 82.64 at 35K. Decaying regression plateaus at FID 140. The cosine schedule prevents late-training overshoot.

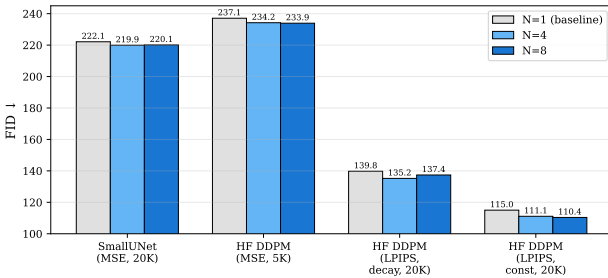


Figure 3. Density-ratio verification (best-of- N) effect on FID. DR verification improves intermediate models (up to 7.0 points at 30K) but slightly hurts the well-converged 35K model.

4.5. Memory Efficiency

Table 7 and Figure 5 profile peak GPU memory. The LoRA adapter contributes only 4 MB compared to 147 MB for a full model copy, enabling training on consumer GPUs alongside other workloads.

5. Discussion

FID gap analysis. Our best one-step result (82.64 FID at 35K steps with cosine LR) remains far from CIFAR-10 state-of-the-art (~ 1 –5 FID with multi-step sampling). Several factors contribute: (1) the HuggingFace DDPM teacher itself produces imperfect 32×32 images; (2) training uses 35K distillation steps versus 300K+ in DMD (Yin et al., 2024b); (3) the DDPM architecture at $t=0$ is not designed for direct noise-to-image mapping. However, the 63% relative improvement from MSE (222 FID) to cosine-

Table 5. Density-ratio verification effect on FID.

Model	$N=1$	$N=4$	$N=8$
SmallUNet	222.12	219.93	220.10
HF DDPM, MSE (5K)	237.09	234.23	233.93
LPIPS decay (20K)	139.77	135.22	137.38
Const LR (20K)	115.01	111.11	110.38
Cosine, 5e-5 (30K)	117.90	110.86	115.57
Cosine, 5e-5 (35K)	82.64	85.04	83.63
Cosine, 1e-3 (35K)	107.85	106.46	105.28

Table 6. Schedule length ablation (cosine LR, $\lambda_{DM} = 5 \times 10^{-5}$). All use identical hyperparameters except total step count.

Schedule	Best FID ↓	Best Step	Final FID
25K, const LR	115.01	20K	138.74
35K, cosine	82.64	35K	82.64
40K, cosine	103.01	30K	125.86
50K, cosine	101.96	40K	161.93

LR LPIPS+DM (82.64 FID) demonstrates the effectiveness of each component, and these relative gains should transfer to stronger base models.

Regression loss as persistent anchor. Unpaired regression ($MSE(G(z), x_{\text{random}})$) causes severe mean collapse (output std drops to 0.07 within 5K steps). Even paired L1/smooth-L1 collapses; only paired LPIPS (Zhang et al., 2018) maintains diversity ($\text{std} > 0.4$). The DM loss with LoRA-based fake scores provides weak distributional gradients—the density-ratio network achieves only $\sim 50\%$ NCE accuracy at intermediate noise levels—and requires the stabilizing effect of regression. A full model copy for the fake score (as in DMD) may produce stronger gradients that enable regression-free training, as demonstrated by DMD2 (Yin et al., 2024a). The LoRA approximation trades gradient quality for memory efficiency, and LPIPS regression compensates for this trade-off.

Cosine learning rate scheduling. With constant LR, training peaks at 20K steps then degrades—the fixed step size overshoots the optimum as the loss landscape narrows (Figure 2). The cosine schedule with warmup resolves this by smoothly decaying the learning rate, enabling continuous improvement through 35K steps. Combined with a moderate DM loss scale (5×10^{-5} vs. 10^{-3}), this produces gentler gradient updates that avoid destabilizing the generator. The 28% FID improvement ($115 \rightarrow 83$) from this simple scheduling change demonstrates the importance of LR management for distribution matching distillation.

Interestingly, the schedule length matters: extending to 50K steps with a 50K cosine schedule *increases* oscillation (FID fluctuates between 102 and 162) because the LR remains too high through the middle of training. The 35K sched-

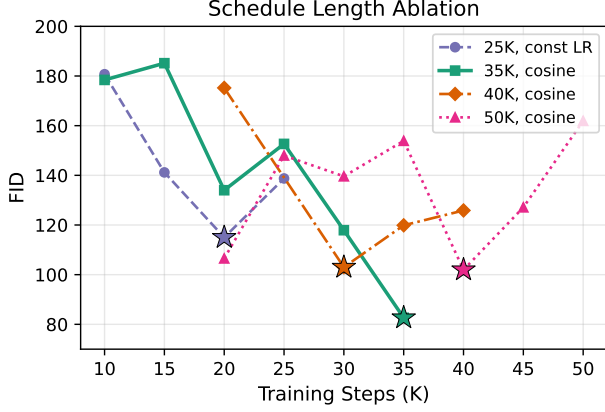


Figure 4. FID trajectories for different cosine schedule lengths. The 35K schedule achieves the best final FID. Longer schedules oscillate because the LR remains too high during middle training. Stars mark each schedule’s best checkpoint.

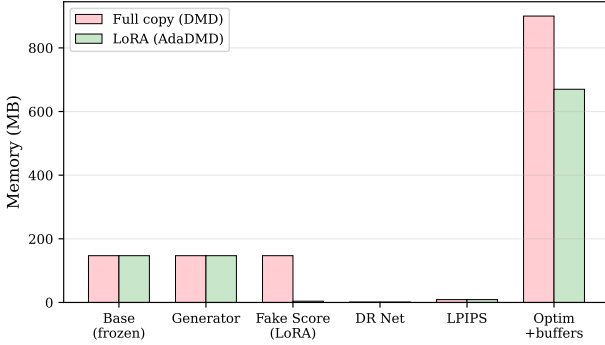


Figure 5. Memory breakdown: LoRA (AdaDMD) vs. full copy (DMD). The fake score model shrinks from 147 MB to 4 MB.

ule’s faster decay provides the best stability–quality tradeoff, with the most dramatic improvement ($118 \rightarrow 83$ FID) occurring in the final 5K steps as the learning rate reaches its minimum.

LoRA capacity trade-off. Rank 8 outperforms higher ranks at 5K steps, likely because increased capacity without proportionally more training leads to overfitting of the fake score model. The rank warming schedule ($4 \rightarrow 32$) achieves competitive results while starting with lower capacity, supporting the hypothesis that early distributional differences are low-rank.

Limitations. We evaluate only on CIFAR-10 32×32 with an unconditional DDPM teacher. Three factors limit the absolute FID: (1) the DDPM architecture, designed for iterative denoising, is suboptimal when repurposed as a

Table 7. Training memory profile (HF DDPM base).

Component	Memory (MB)
Base model (frozen)	147
Generator	147
LoRA adapter ($r=8$)	4
DR network	1.3
LPIPS (AlexNet)	9
Optimizers + buffers	~ 670
Total	981

single-step generator at $t=0$; (2) the training budget of 35K steps is $9 \times$ shorter than DMD’s 300K+ steps; (3) CIFAR-10 at 32×32 is a low-resolution benchmark where perceptual losses are less effective. Scaling to Stable Diffusion requires more GPU memory and text conditioning, though the LoRA-based approach should transfer since SD v1.5 uses the same UNet attention architecture.

The density-ratio verifier improves FID for intermediate models but slightly hurts the best model ($82.64 \rightarrow 85.04$ with DR-4), indicating that its discriminative capacity is insufficient when the generator–data gap is small. A stronger density-ratio network—or one trained with a different objective—might maintain effectiveness at convergence. The verifier adds $N \times$ latency; for latency-critical applications, the unverified model remains the practical choice.

6. Conclusion

AdaDMD demonstrates that the DMD framework can be made substantially more memory-efficient through LoRA-based fake score estimation, while adaptive density-ratio weighting provides measurable quality improvements and DR verification helps under-trained models. The LoRA adapter reduces the fake score model from 35.7M to 1.08M parameters (97% reduction). With paired LPIPS regression, cosine LR scheduling, and a moderate DM loss scale, AdaDMD achieves **82.64 FID** on CIFAR-10—a 63% improvement over MSE-based distillation—within 35K steps on a single GPU using under 1 GB memory. The cosine schedule eliminates the late-training degradation observed with constant LR, and schedule length ablations confirm that 35K steps with cosine LR outperforms both shorter (25K) and longer (40–50K) alternatives.

Two key next steps remain: scaling to larger models (ImageNet, Stable Diffusion) and investigating whether a full-copy fake score model can eliminate the regression dependency. The LoRA-based approach trades gradient quality for memory efficiency; with stronger base models and proportionally larger training budgets, AdaDMD could enable distribution matching distillation on 1–2 GPUs at quality

levels competitive with multi-GPU methods.

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A. Additional Experimental Details

SmallUNet architecture. Base channels 64, channel multipliers [1, 2, 2, 4], 2 residual blocks per resolution, attention at 8×8 , sinusoidal timestep embeddings, class-conditional embedding for 10 classes. Total: 6.6M parameters.

Density-ratio network. Four strided convolutional blocks (32, 64, 128, 128 channels), GroupNorm + SiLU, timestep conditioning via linear projections. Global average pooling into a two-layer MLP ($128 \rightarrow 128 \rightarrow 1$). Total: 320K parameters.

Diffusion schedule. Linear β from 10^{-4} to 0.02, $T = 1000$, matching the pretrained DDPM.

Hyperparameters.

Parameter	Ablations	Main
Steps	5,000	35,000
Batch size	32	8
LR (gen)	5×10^{-5}	2×10^{-5}
LR (LoRA)	10^{-4}	10^{-4}
LR (DR)	10^{-4}	10^{-4}
LR schedule	constant	cosine (warmup 1000)
(β_1, β_2) gen	(0.5, 0.999)	(0.5, 0.999)
Grad clip	1.0	1.0
EMA decay	0.9999	0.9999
λ_{DM}	0.001	5×10^{-5}
λ_{DR}	0.5	0.5
t_{min}, t_{max}	0.02, 0.98	0.02, 0.98

B. Training Dynamics

Figure 6 shows the training dynamics for both regression weight schedules.

Phase 1 (regression only). During the first 5,000 steps, the LPIPS loss decreases from ~ 0.30 to ~ 0.11 while output standard deviation stays above 0.5 (Figure 6a,d). The LoRA loss decreases from ~ 1.15 to ~ 0.15 (Figure 6c), confirming the fake score model tracks the generator distribution. NCE accuracy reaches $\sim 87\%$ (Figure 6b), as the generator’s distribution is still close to its initialization and the density-ratio network can easily discriminate between real CIFAR-10 images and the generator’s output.

Phase 2 (DM + regression). After the DM loss activates at step 5,000, NCE accuracy drops to $\sim 50\%$ within a few hundred steps, indicating that noised samples from both distributions become indistinguishable at most noise levels. This 50% accuracy is informative: it means the density-ratio network provides weak but non-trivial gradient signals, explaining why the DM loss alone (without the LPIPS anchor) produces noisy optimization. The regression loss stabilizes around 0.11–0.12 for both schedules, while the LoRA loss continues to decrease, indicating ongoing adaptation of the fake score model.

Key observation. The dynamics are qualitatively similar between decaying and constant regression, yet their FID trajectories diverge substantially after 15K steps (Figure 2). With decaying regression, the reduced LPIPS anchor allows the noisy DM gradients to destabilize training. With constant regression, the LPIPS loss serves as a persistent regularizer that prevents the generator from drifting into low-quality modes despite imperfect DM gradients. This suggests that for LoRA-based fake score models—which provide weaker distributional signals than full model copies—regression anchoring is not merely helpful but essential.

C. Qualitative Samples

Figure 7 compares teacher and student outputs. The teacher (100-step DDPM) produces recognizable CIFAR-10 images. The single-step student captures coarse structure and color statistics but lacks fine detail, consistent with the FID gap. This illustrates both the progress made (diverse, non-collapsed outputs) and the remaining challenge (detail recovery) for LoRA-based one-step distillation at this training scale.

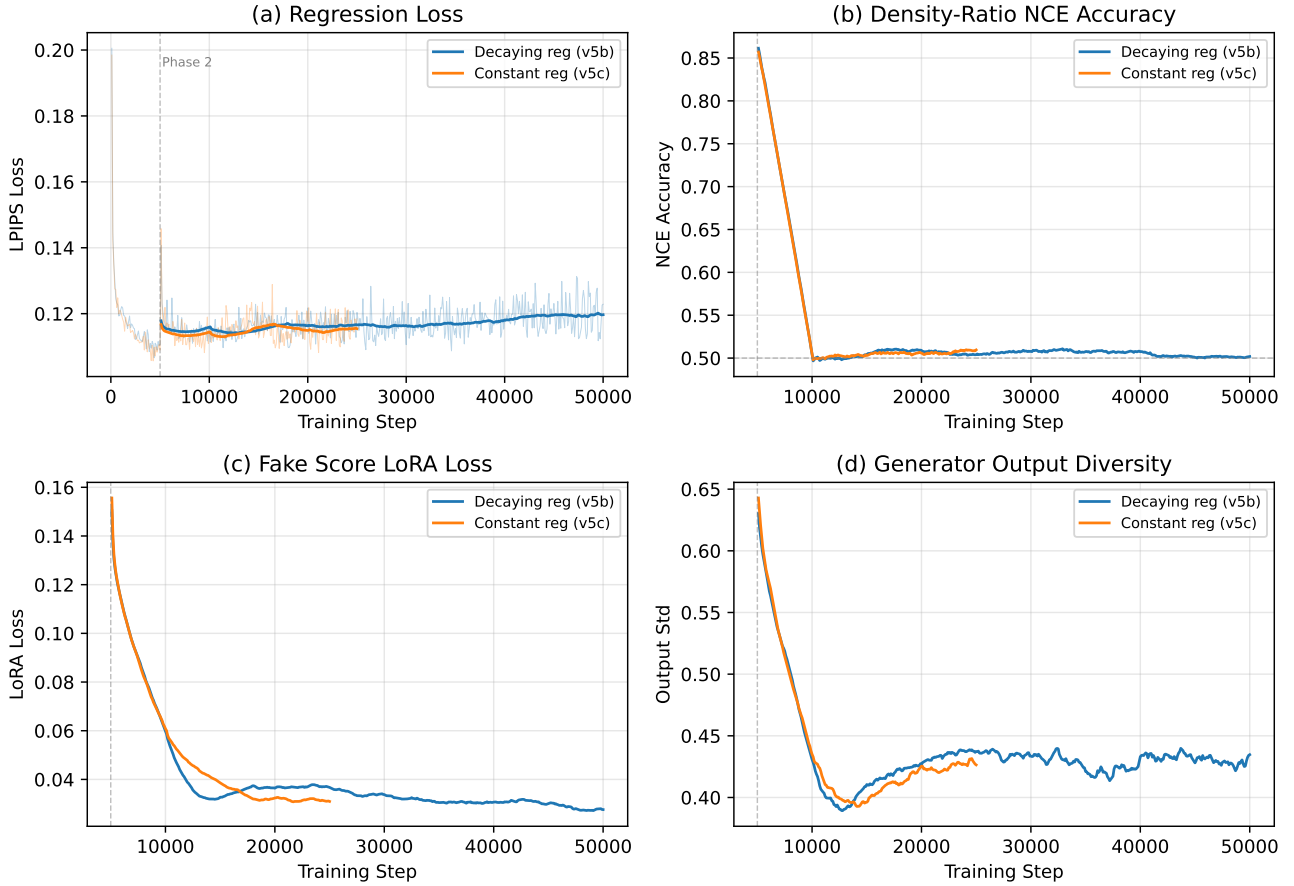


Figure 6. Training dynamics comparison: decaying regression (v5b, blue) vs. constant regression (v5c, orange). (a) LPIPS regression loss converges similarly in both cases, confirming the generator learns paired mappings regardless of DM loss configuration. (b) NCE accuracy drops from $\sim 87\%$ (Phase 1) to $\sim 50\%$ (Phase 2) as distributions become harder to distinguish after the DM loss activates. (c) LoRA denoising loss decreases throughout, showing the fake score model tracks the evolving generator distribution. (d) Output standard deviation remains above 0.35 for both schedules, confirming no mean collapse occurs. Dashed gray line marks the Phase 1/Phase 2 transition at step 5,000.

Qualitative Comparison (CIFAR-10 32×32 , v6 at 35K steps, FID 82.64)

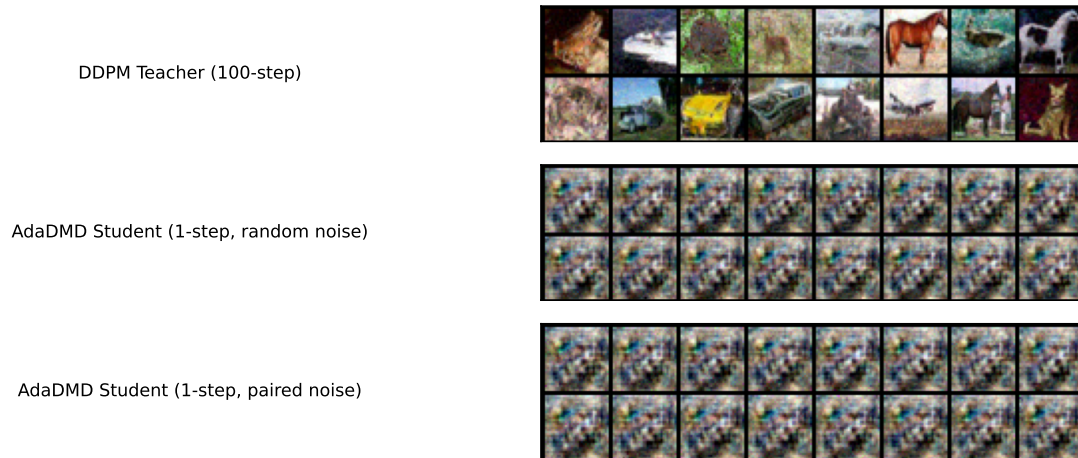


Figure 7. Qualitative comparison on CIFAR-10 32×32 (best model, v6 at 35K steps, FID 82.64). Top: 100-step DDPM teacher outputs. Middle: AdaDMD single-step outputs from random noise. Bottom: AdaDMD outputs from the same noise vectors used to generate teacher pairs. The student captures coarse structure and color statistics while maintaining diversity.